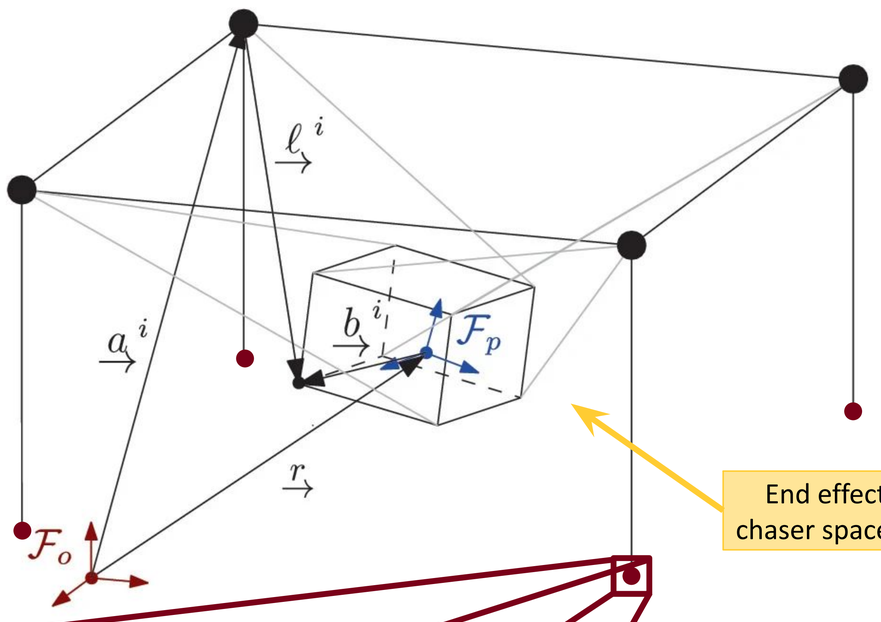
Spacecraft Motion Simulation Using Cable-Driven Parallel Robots

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Cable-Driven Parallel Robots (CDPRs): Motivation and State of the Art



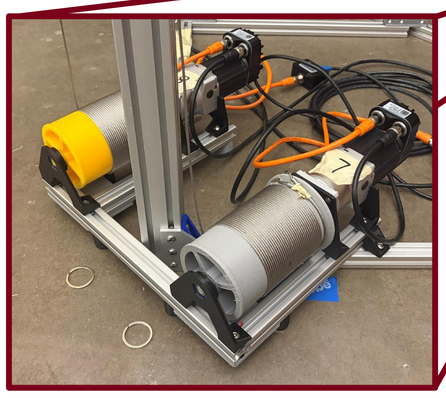
CDPRs

- End effector controlled via eight cables, actuated by winches.
- Example: Skycam, a four cable CDPR
- Pros: 6-DOF, lightweight, low inertia, high acceleration, large workspace.
- Cons: cable sag, challenging estimation and forward kinematics.

Skycam

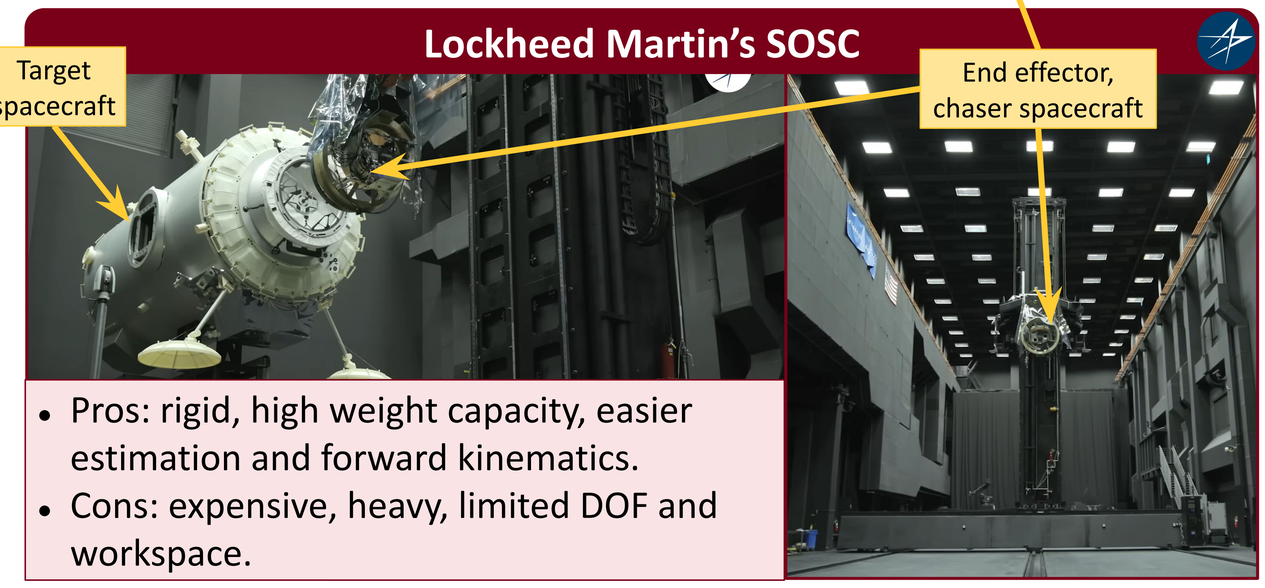


State-of-the-Art Motion Simulator



- Lockheed Martin's Space Operations Simulation Center (SOSC)
  - Serial, linear actuators.
  - Used for Artemis Mission's testing of docking between Orion and Human Landing System.

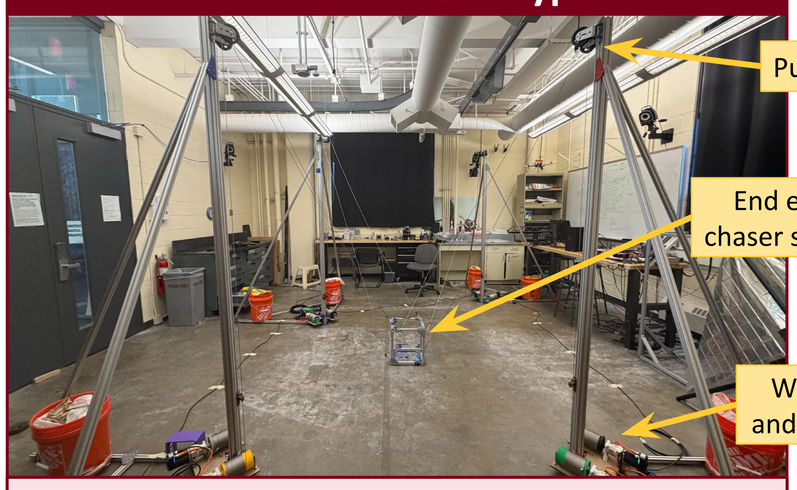
Lockheed Martin's SOSC



- Pros: rigid, high weight capacity, easier estimation and forward kinematics.
- Cons: expensive, heavy, limited DOF and workspace.

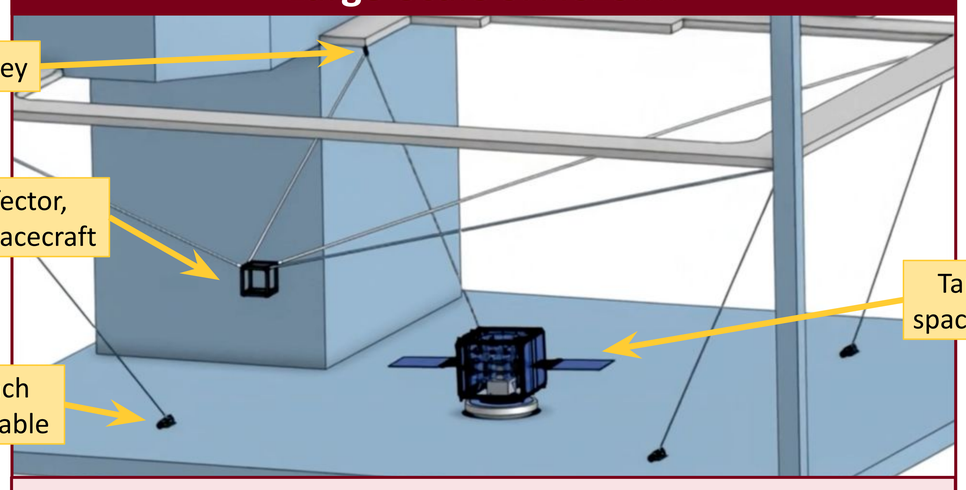
Spacecraft Platform Actuated by Cables for Simulated Motion (SPACESiM)

Small-Scale Prototype



- Initial prototype for CDPR hardware and software testing, and relative motion
- Learning lessons in lower stakes environment before large-scale.

Large-Scale SPACESiM



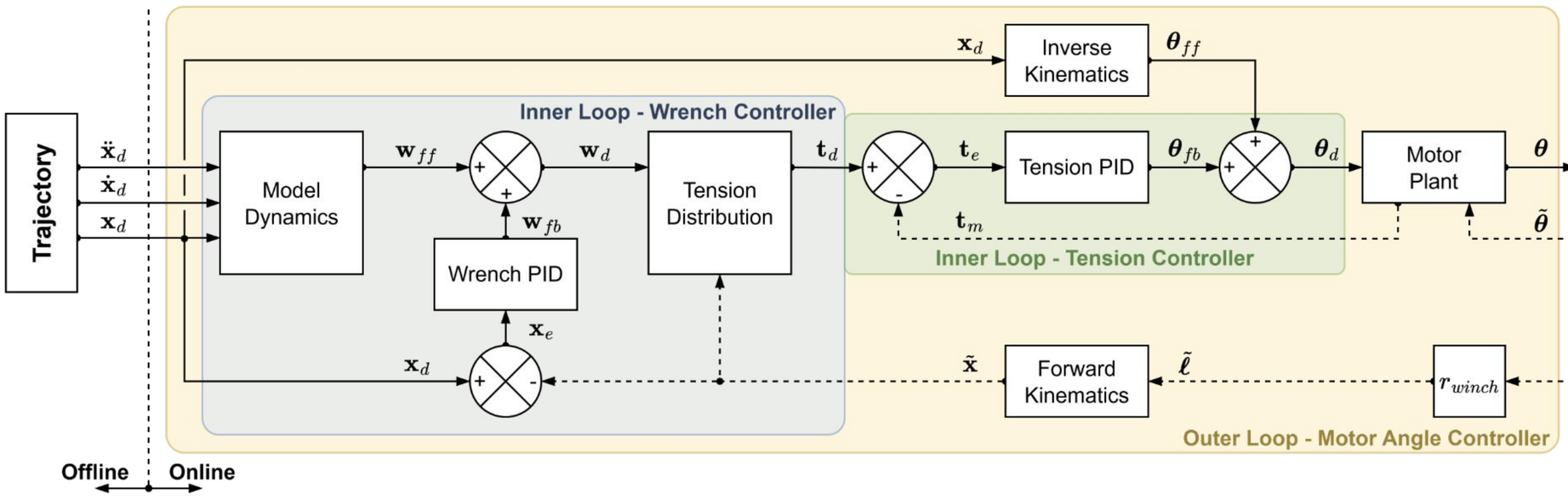
- 40' x 40' x 40' room with 100 lb capacity.
- Hardware and software in the loop testing.
  - Guidance, navigation, and control algorithms; cameras, LiDAR, radar, IMU, and horizon sensors.

Real World Applications



- Rendezvous, proximity operations, and docking.
- On-orbit servicing, assembly, and manufacturing.

Ongoing Work: Control and Estimation Algorithm



- New cascade control algorithm managing:
  - Pose trajectory tracking, cable tension and sag, wrench trajectory tracking.
- Challenges: heavy filtering on tension, real time code at 333 HZ.

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